[AI Cluster]



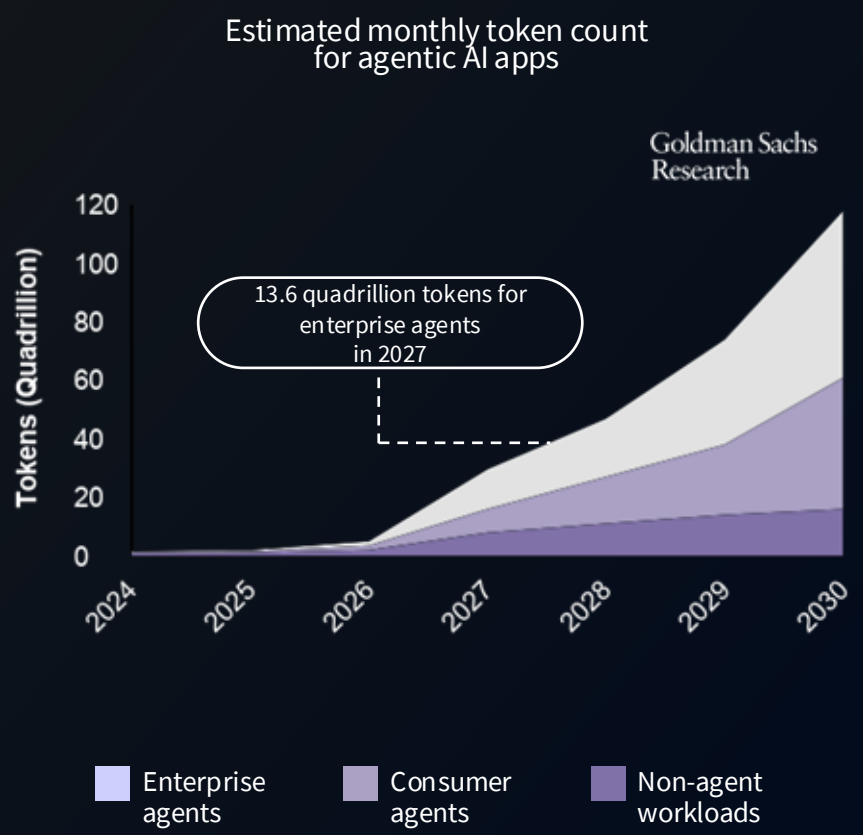
Arm AGI CPU × Rebellions RebelServer: A System Approach to Agentic AI Infrastructure

Dilip Ramachandran, *Sr. Director of Marketing,* ***Arm***

Jennifer Glore, *EVP Product Management,* ***Rebellions***

Agents drive CPU demand

AI agents token usage is expected to rise 24x by 2030



CPU

ACCELERATOR

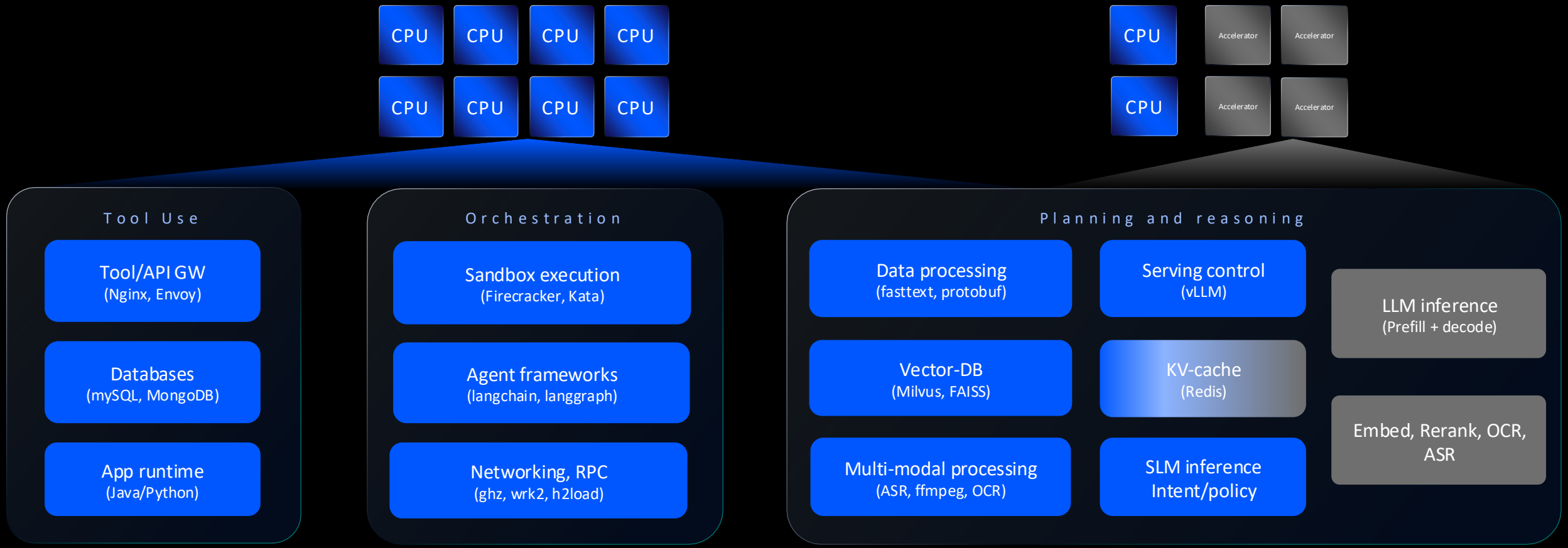
Task distribution

Enterprise automation
Dominated by concurrent interactions
50% | 50%

Software development
Dominated by dependency chains
60% | 40%

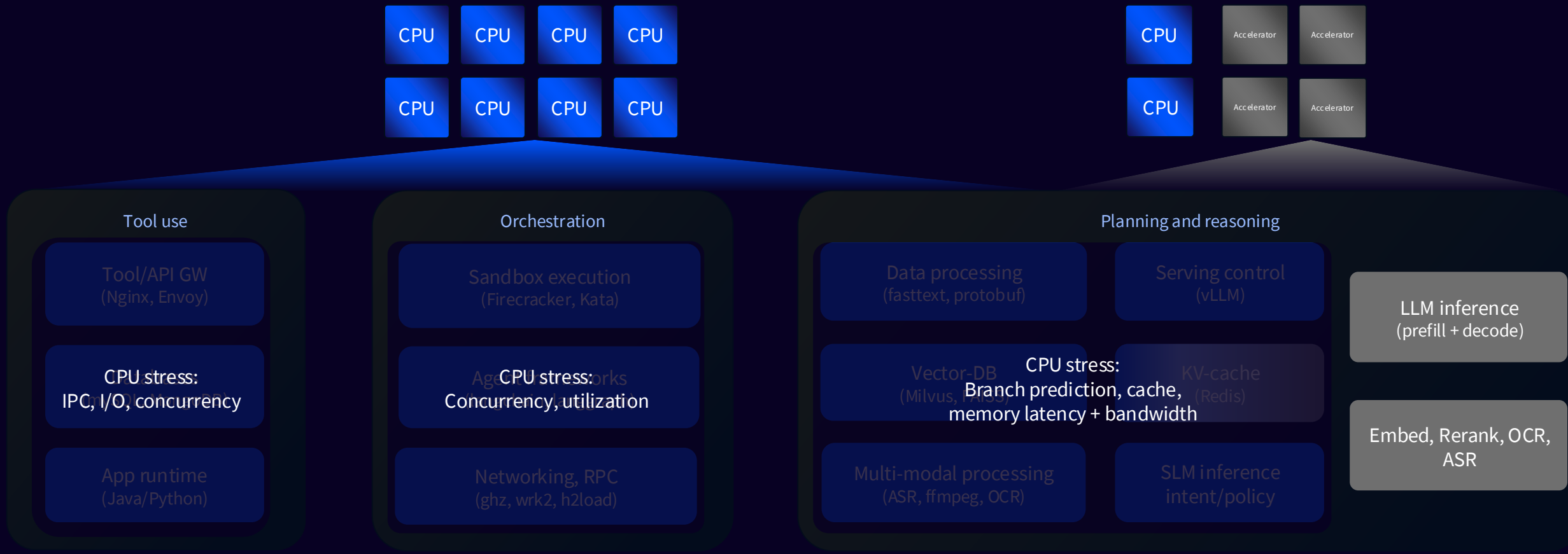
Financial analysis
Dominated by parallel decomposition
40% | 60%

Agentic Workload Performance



© 2026 Arm

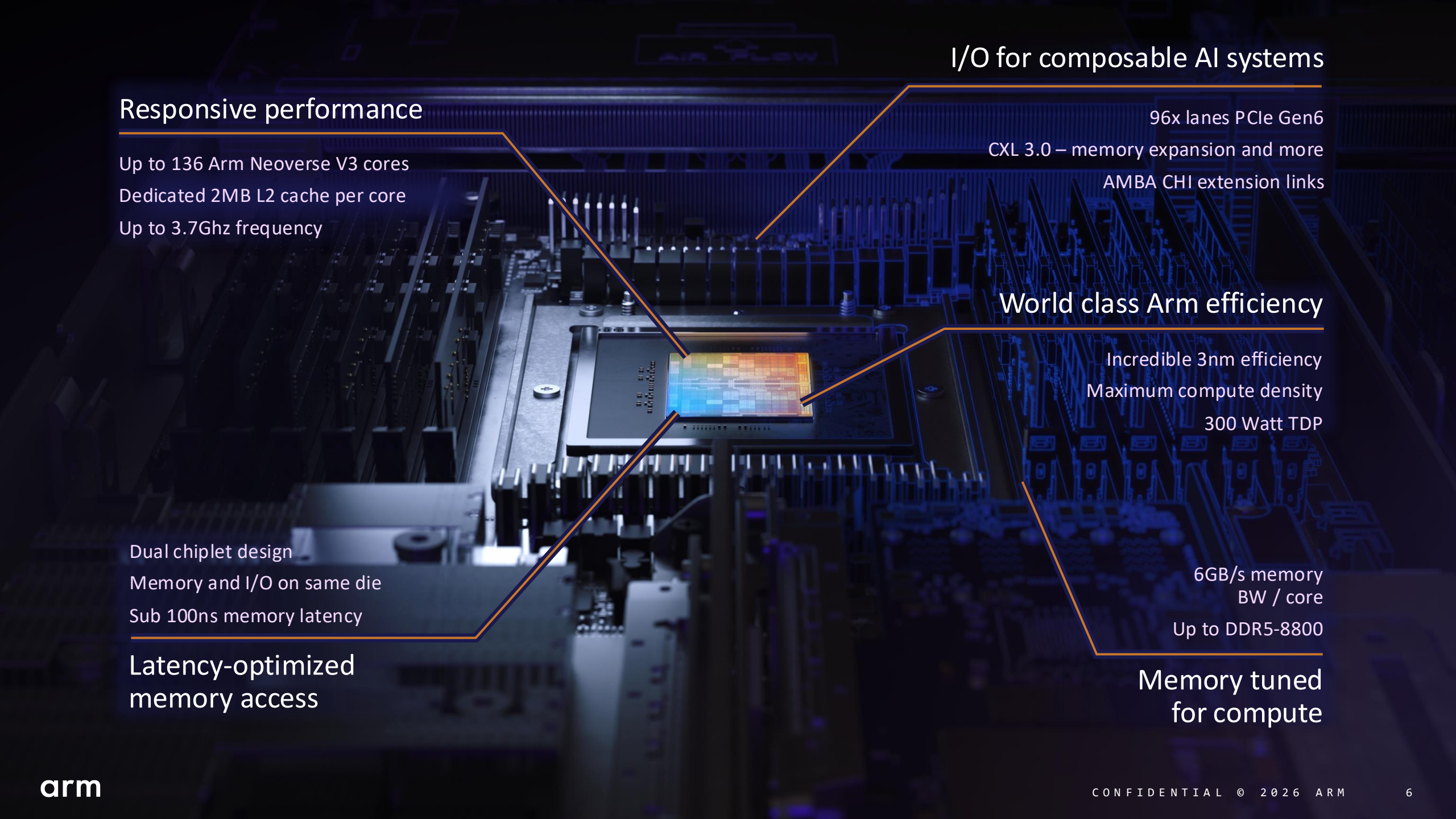
Agentic workload performance



Key agentic performance indicators:

- Worst case (p99) performance at each step (tools, orchestration, planning/reasoning)
- Start-up/switching latency and steady-state throughput


arm AGI CPU
World's most efficient agentic CPU
Deployment Partners
cerebras
CLOUDFLARE
f5
Meta
OpenAI
ORACLE
SAP
SK telecom
verda
火山引擎 VolcanoEngine
OCP GLOBAL SUMMIT 2026
Scaling Innovation for the AI Era.



Responsive performance

Up to 136 Arm Neoverse V3 cores

Dedicated 2MB L2 cache per core

Up to 3.7Ghz frequency

I/O for composable AI systems

96x lanes PCIe Gen6

CXL 3.0 – memory expansion and more

AMBA CHI extension links

World class Arm efficiency

Incredible 3nm efficiency

Maximum compute density

300 Watt TDP

Dual chiplet design

Memory and I/O on same die

Sub 100ns memory latency

Latency-optimized memory access

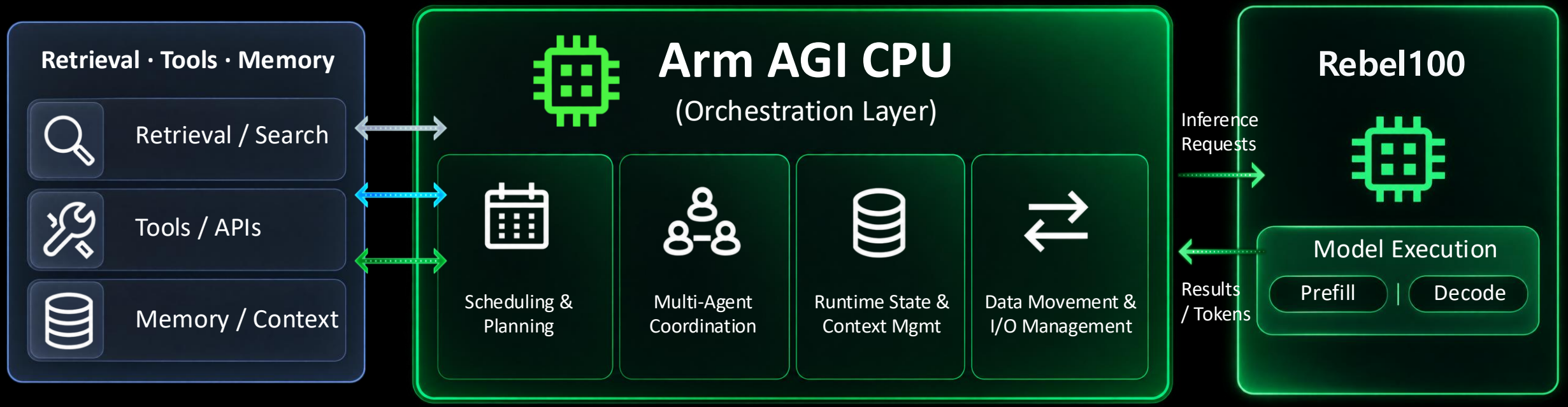
6GB/s memory BW / core

Up to DDR5-8800

Memory tuned for compute

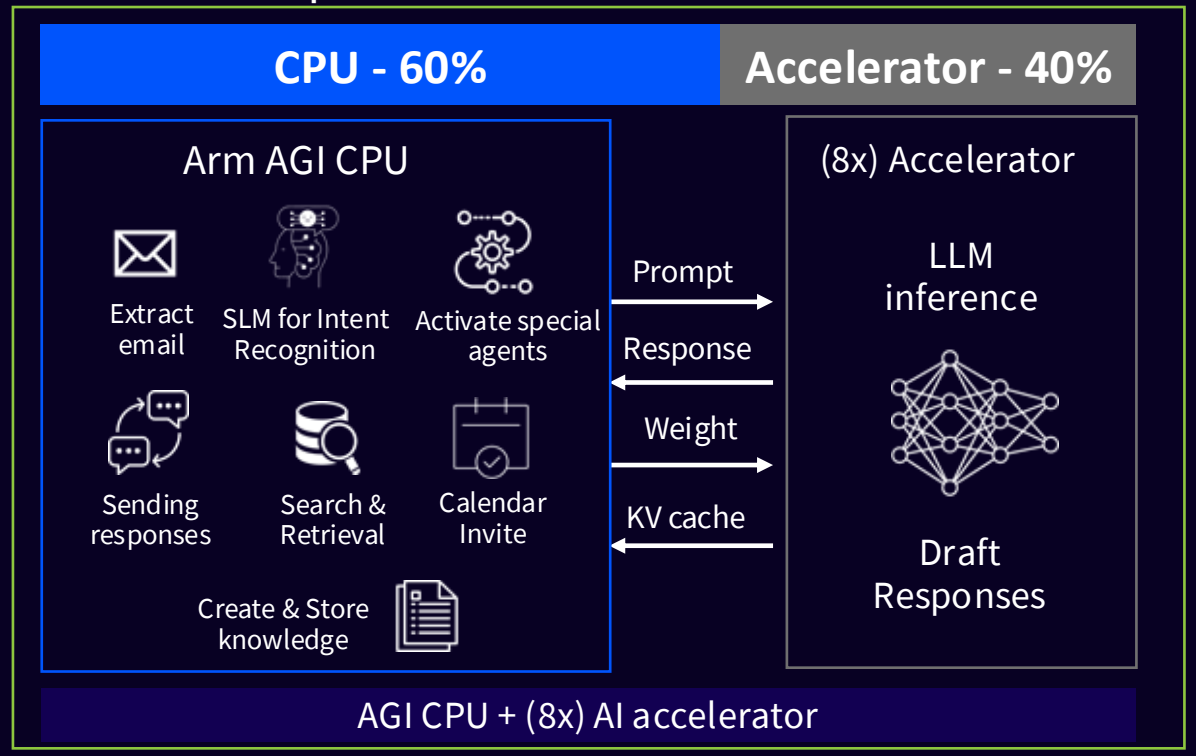
arm

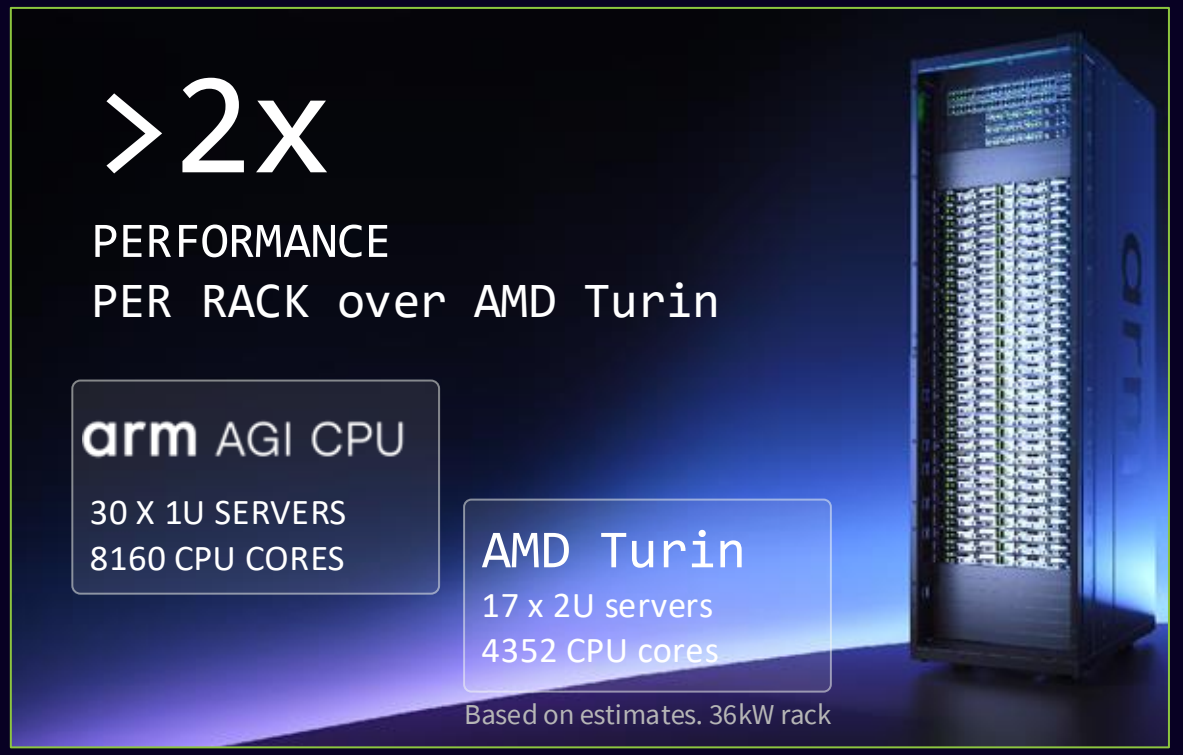
Optimizing Agentic AI Inference
with Arm AGI CPU and Rebel100™



Agentic AI use-case analysis - Customer support

Split between CPU and GPU*





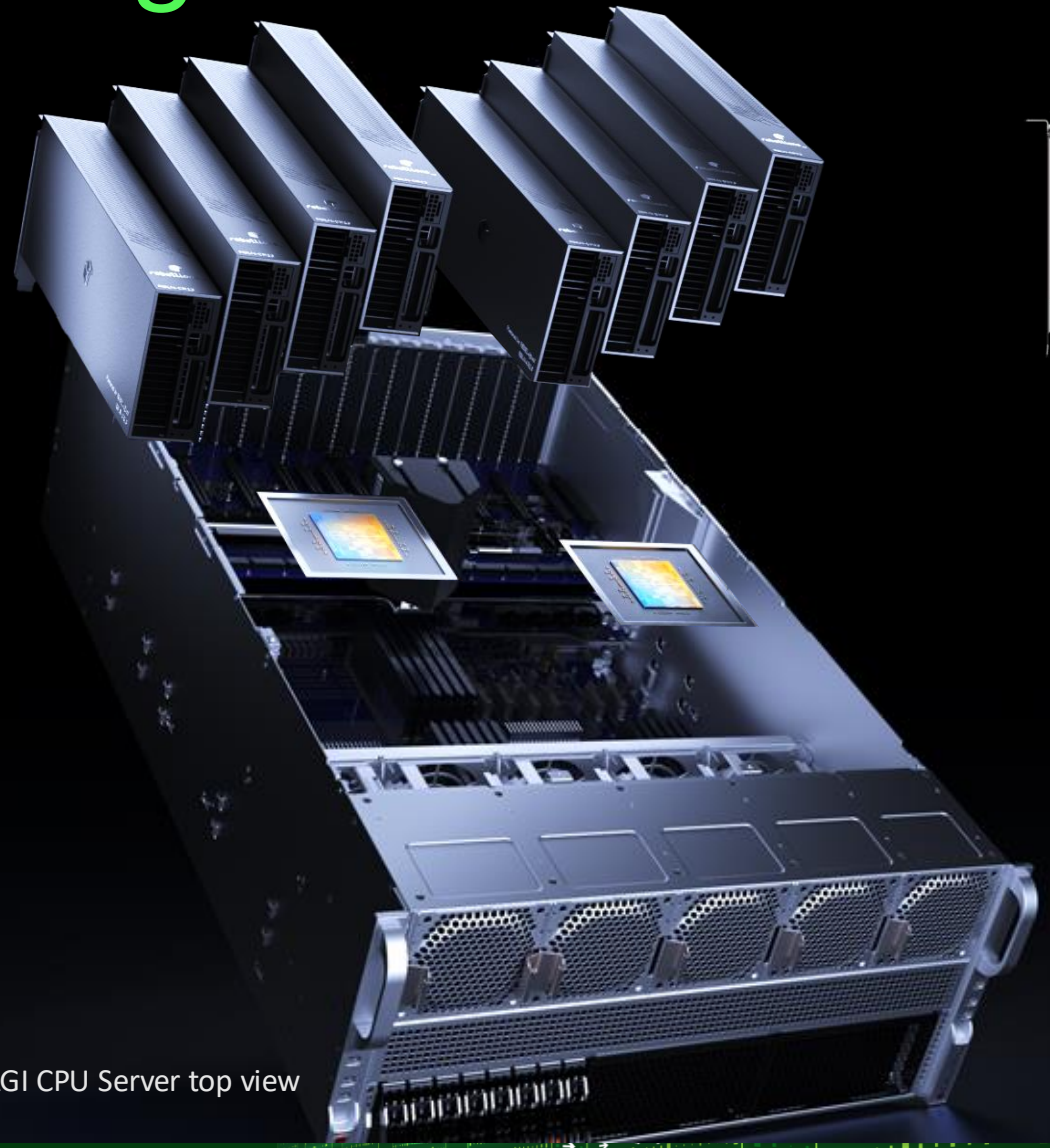
Large % of Agentic AI runs on CPU

* Based on PoC analysis done by Arm

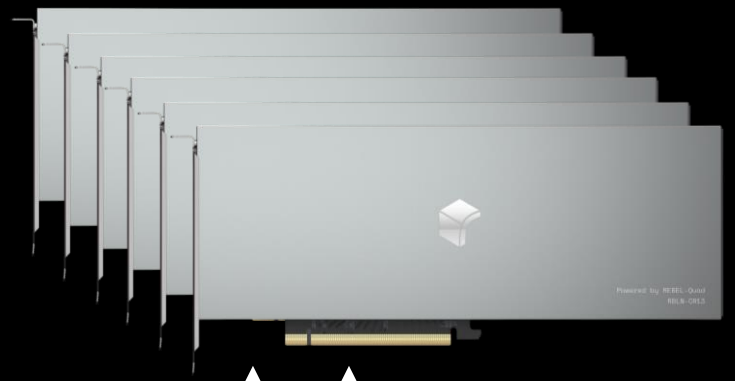
Arm outperforms x86 for Agentic AI

* Based on measurements + projections

Integrated Hardware: RebelServer™



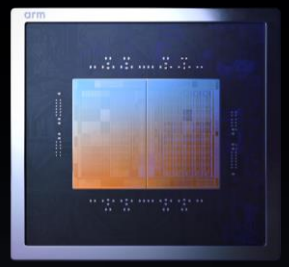
Arm AGI CPU Server top view



rebellions_

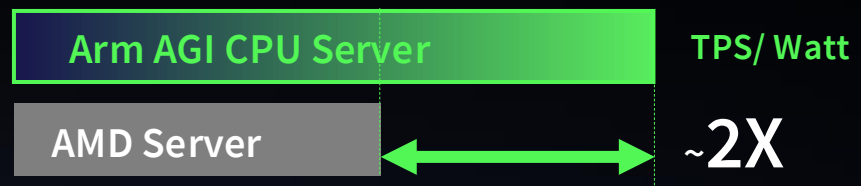
RebelCard™

Seamless Integration for
Efficient AI Inference

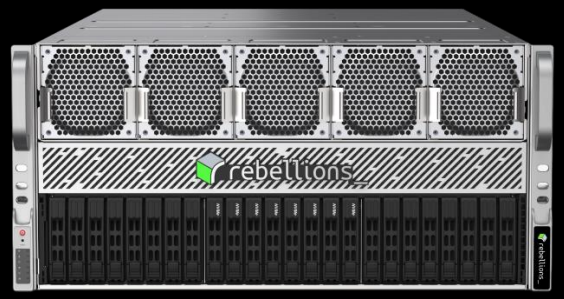


arm

AGI CPU



Scaling from Server to Data Center with Rebel



RebelServer

4x RebelServer per Rack
32x RebelCard™ NPU



RebelRack™

1–7 NPU Racks
4–28 Nodes
High-availability production



RebelPOD™

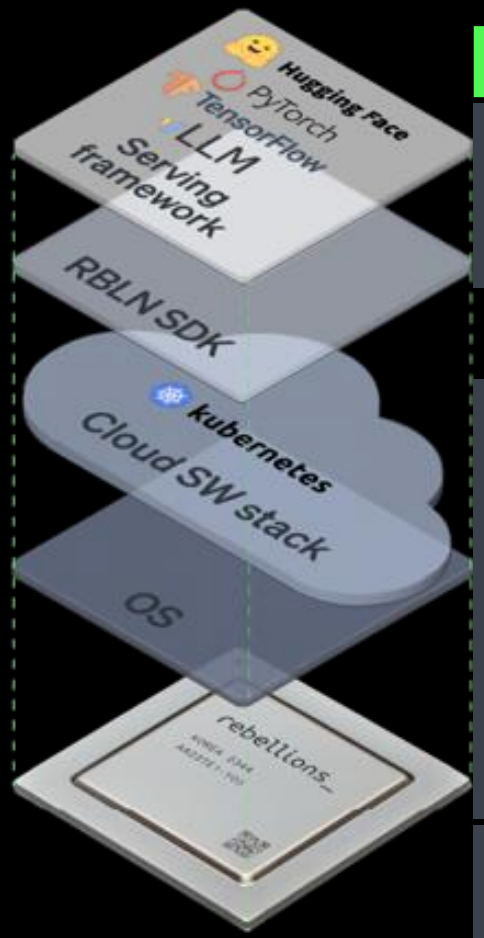
8+ NPU Racks
32+ Nodes
Massive AI factory deployment



Data Center

Multi-POD scale
Sovereign AI infrastructure

Easy to Use Software



HW/SW co-optimized full AI stack

RBLN Cloud Toolkit

NPU operator
Kubernetes plugin, metrics exporter, VFIO manager, device installer, node feature discovery

Compiler

Runtime

Profiler

Model Zoo
compatible with PyTorch, HuggingFace

vllm-rbln
customize/extend vLLM features for NPU

torch-rbln
Natively support PyTorch without model porting

System management tools (rbln-smi, etc)
Kernel driver
Firmware

Complete integration with existing AI workflows

ready for deployment

vLLM official plugin

PyTorch out-of-tree extension

Cloud Platform

openstack.
kubernetes
Prometheus Grafana ceph Istio

AI Stack

PyTorch Hugging Face
vLLM python
C C++

OS Support

RHEL 9.4. Ubuntu 22.04/24.04

[Call to Action]

Explore the Arm + Rebellions AI Inference Solution

01 Explore the Solution

Combining Arm AGI CPU orchestration with Rebellions Rebel100 to power agentic AI workloads, from server to data center scale.

arm

CPU orchestration & workload management

rebellions_

AI inference acceleration & scalable infrastructure

02 Learn More

Explore our technologies and find detailed information.

Arm

Booth C72 - Agentic Demos

https://www.arm.com/products/cloud-datacenter/arm-agi-cpu

Rebellions

Booth S10

https://rebellions.ai/

03 Get Involved

Connect with Arm and Rebellions to explore deployment opportunities, collaboration, and future projects.

Join the community and explore deployment opportunities
Let's start the conversation

Ready to scale agentic AI inference?
EXPLORE ARM AGI CPU + REBELLIONS REBEL100


Thank
You!
OCP
GLOBAL
SUMMIT
October 12-15, 2026
San Jose, California